

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • GPA: 3.75/4.0

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

EXPERIENCE

Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Architected event-driven **TypeScript/React** system with observer pattern decoupling UI state propagation – accelerated feature delivery **68%** across **25K+** lines of Chromium at **95% test coverage**
- Designed **C++ IPC transport layer** with Protocol Buffers – selected Protobuf over custom serialization for schema evolution and cross-language compatibility; shipped to Chrome stable serving **3B+** users at **sub-50ms p99, 10K+ req/sec**
- Identified settings navigation as p99 bottleneck at 1,200ms; implemented **lock-free concurrent trie search** eliminating mutex contention – **96% latency reduction**, zero production regressions
- Delivered design documents and code reviews to the Chrome infrastructure team; changes adopted into the production branch reviewed by senior Chrome engineers

TiMoto AI | Software Engineer

Sep 2025 – Present

- Primary engineer for backend, cloud infrastructure, and ML serving on a 3-person team – designed, deployed, and operated **distributed production systems** end-to-end across **Django, PostgreSQL, and gRPC**
- Built **gRPC inter-service layer** over a PostgreSQL data model; resolved production deadlock under concurrent calls by tracing shared resource acquisition and redesigning call sequencing – **100% evaluation success** at **sub-50ms p99**
- Architected **vLLM inference engine with PagedAttention** for LLM serving – eliminated KV cache fragmentation under concurrent load with continuous batching, achieving **zero OOM failures** at production traffic
- Architected multi-AZ ECS Fargate with **Terraform IaC**, CloudWatch observability, and circuit breakers – **99.9% uptime, 44% cost reduction**; auto-rollback on health check failure and on-call rotation

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on AWS – selected JWT over session auth for horizontal scalability; auto-scaling for **500+ concurrent users, 90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned with **PostgreSQL indexing** achieving **sub-100ms** for 10,000+ records

PROJECTS

Pulumi – Contributor | Go, TypeScript, IaC

github.com/pulumi

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning – under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization – studied linearizability guarantees and correctness under concurrent operations and partial failures

SKILLS

Languages: TypeScript, JavaScript, Python, Go, C++, Java, SQL, Bash, Rust

Web & Frameworks: React, Node.js, Django, FastAPI, REST APIs, HTML, CSS, Redux

Databases: PostgreSQL, Redis, MongoDB, Elasticsearch-adjacent search/indexing

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, K8s, Docker, CI/CD, CloudWatch

Distributed & ML Systems: gRPC, Protocol Buffers, Circuit breakers, Event-driven architecture, vLLM, PagedAttention

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor